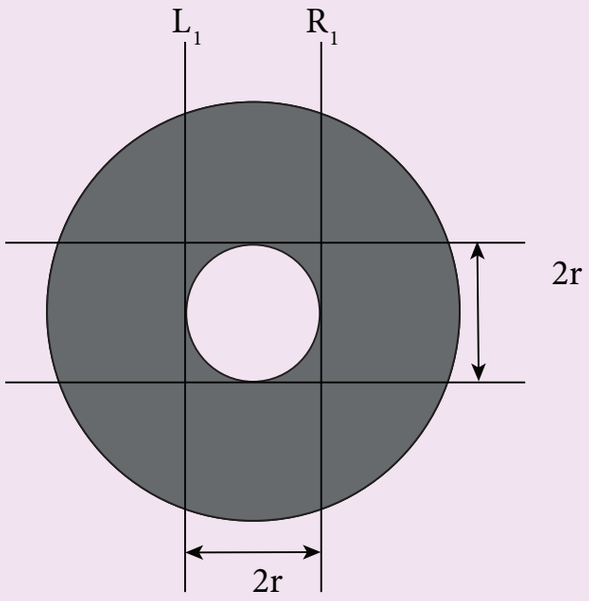
- Fix a needle near the capillary tube so that the needle touches the water surface
- A Vernier microscope is focused at the lower meniscus of the water and the corresponding reading is taken after coinciding it with the horizontal line of the cross wire.
- Tip of the needle is focussed using vernier microscope after coinciding it with horizontal line of the cross wire
- The difference between the two readings of the vertical scale gives the height (h) of the liquid raised in the capillary tube.
- Now to find the radius of the tube, raise the capillary tube and remove the beaker. Carefully rotate the capillary tube so that the immersed lower end face towards you.
- Focus the capillary tube using Vernier microscope to clearly see the inner walls of the tube.
- Let the vertical cross wire coincide with the left side inner walls of the tube. Note down the reading $(L_1)$
- Turn the microscope screws in horizontal direction to view the right side inner wall of the tube. Note the reading $(R_1)$. Thus the radius of the tube can be calculated as $\frac{1}{2}(L_1 - R_1)$.
- Finally calculate the surface tension using the given formula.




RADIUS OF THE CAPILLARY TUBE

## OBSERVATIONS

To measure height of the liquid (h)

Least count of the microscope = _____ *cm*



UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 298 17-08-2020 16:26:43